

Forecast-available signals

311 history + calendar
Weather at t + IDs



Leakage-controlled panel

NTA $\times$ Week $\times$ Category
History at $t \rightarrow$ Event at $t+1$

$t \rightarrow t+1$ design



Prospective evaluation

LightGBM + XGBoost
2015-22 Train | 2023 Val.
2024-25 Test



Category-calibrated alerts

Ensemble score $\rightarrow$
service threshold $\rightarrow$
risk alert
SHAP review

validation only

OSM + PLUTO 2026: retrospective context only

excluded from prospective alerts